SAPPHIRE LANGUAGE SPECIFICATION AND AUTOMATION MANUAL

Formal specification of Sapphire v1.0.0 language semantics, type system, standard library APIs, and the ml.distributed frontier training engine.

1. Language Primitives

Types: string, int, float, bool, list, dict, nil

Keywords: let, fn, if, else, for, in, while, return, parallel, import, print

Operators: + - * / % == != < > <= >= && || ! |>

2. Standard Library Namespaces

Namespace	Key APIs
os	system_info(), notify(), clip_write(), clip_read(), sleep()
fs	read(), write(), exists(), delete(), list_dir(), copy()
http	get(), post(), put(), delete(), download()
ai	prompt(), classify(), extract_json(), embed(), summarize()
ml	tensor(), randn(), rand(), zeros(), matmul(), autograd, model, train, gpu, kernel, distributed
ml.distributed	Transformer(), Cluster(), train() -- 5D frontier LLM engine
agent	memory.remember(), .recall(), autonomy.run_loop(), tools.register()
scheduler	interval(), once(), cron(), cancel()
gui	alert(), dialog(), window(), progress_bar()
data	json.parse(), json.stringify(), csv.read(), yaml.parse()

3. ml.distributed.Transformer(config) -- Parameters

Parameter	Default	Description
layers	32	Number of transformer decoder layers
hidden	4096	Hidden dimension size (d_model)
heads	32	Number of attention heads
kv_heads	None	GQA: number of KV heads (None = MHA)
ff_mult	4	FFN size = ff_mult * hidden (SwiGLU)
vocab	32000	Vocabulary size
seq_len	4096	Maximum sequence length
precision	bf16	Training precision: bf16 or fp8
moe_experts	0	MoE: number of experts (0 = dense)
top_k	2	MoE: top-k expert routing

4. ml.distributed.train() -- Result Fields

Field	Description
result.strategy	Strategy string e.g. TP1_PP1_DP512_EP1_SP1_ZeRO3_FP8
result.mfu_percent	Model FLOPs Utilization in %
result.tokens_per_sec	Training throughput in tokens/second
result.eta_days	Estimated days to complete total token budget
result.memory_per_gpu_gb	Peak memory used per GPU in GB
result.nccl_latency_ms	Estimated AllReduce latency in milliseconds

result.codegen_path	Path to generated PyTorch FSDP + torchrun script

5. Kernel Dispatch Table (H100-80GB)

Operation	Kernel	Notes
Attention (seq<=8192)	FlashAttention-3	SRAM-only, IO-optimal, H100/H200 only
Attention (seq>8192)	Triton Ring Attention	Distributed long-sequence attention
MLP SwiGLU	Triton Fused SwiGLU	Fused gate + MLP in single kernel
LayerNorm/RMSNorm	Triton Fused Norm	Backward + forward in one pass
RoPE Embeddings	CUDA Fused RoPE	In-place position encoding
Matmul (FP8)	TransformerEngine cublas	E4M3/E5M2 with BF16 accumulation
Matmul (BF16)	cuBLAS Tensor Cores	Standard BF16 GEMM
AllReduce	NCCL Ring	Topology-aware, async overlap with backward

6. 5D Auto-Parallelism Solver Algorithm

Grid search over: TP in {1,2,4,8} x PP in {1,2,4,8} x DP = total_gpus/(TP*PP)
x EP in {1,4,8,64} x SP in {1,2,4} x ZeRO in {1,2,3}

Feasibility filters:

1. TP*PP*DP must equal num_gpus
2. Per-GPU memory (weights+grads+optimizer+activations) <= gpu_vram*0.95
3. PP>1 requires layers % PP == 0
4. TP>1 requires heads % TP == 0

Objective: maximize MFU = compute_flops / (wall_time * peak_flops * num_gpus)
NCCL AllReduce latency is subtracted from compute time in the objective.